

第八章 多元函数积分学 (数一专题)

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考点1: 三重积分

球面坐标 / 轮换对称性

- 设 $\Omega = \{(x, y, z) | x^2 + y^2 + z^2 \leq 1\}$, 则 $\iiint_{\Omega} \underbrace{z^2}_{= r^2 \cos^2 \varphi} dx dy dz = \underline{\hspace{2cm}}.$

解 球面坐标系下, $\Omega = \{(r, \varphi, \theta) | 0 \leq r \leq 1, 0 \leq \varphi \leq \pi, 0 \leq \theta \leq 2\pi\}.$

$$\begin{aligned}\iiint_{\Omega} z^2 dv &= \frac{1}{6} \iiint_{\Omega} \underbrace{(x^2 + y^2 + z^2)}_{r^2} dv = \frac{1}{6} \iiint_{\Omega} r^2 \cdot r^2 \sin \varphi dr d\varphi d\theta \\&= \frac{1}{6} \int_0^{2\pi} d\theta \int_0^{\pi} \sin \varphi d\varphi \int_0^1 r^4 dr \\&= \frac{1}{6} \times 2\pi \times 2 \times \frac{1}{6} = \frac{\pi}{9}.\end{aligned}$$

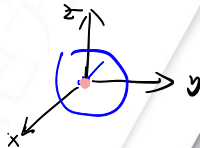
中值定理.

- 设 f 为连续函数, $f(0,0,0) = 1$, Ω 为球心在原点, 半径为 r 的球体, 则极限 $\lim_{r \rightarrow 0} \frac{1}{\pi r^3} \iiint_{\Omega} f(x,y,z) dv = \underline{\hspace{2cm}}$.

解: $\iiint_{\Omega} f(x,y,z) dv = f(\xi, \eta, \zeta) \cdot \frac{4}{3}\pi r^3$ <中值定理>
 $(\xi, \eta, \zeta) \in \Omega$

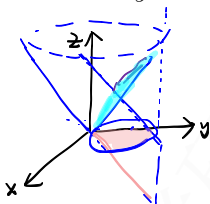
$(\xi, \eta, \zeta) \rightarrow (0,0,0) \xrightarrow{f \text{ 连续}} f(\xi, \eta, \zeta) \rightarrow f(0,0,0)$

$$\lim_{r \rightarrow 0} \frac{\iiint_{\Omega} f(x,y,z) dv}{\pi r^3} = \lim_{r \rightarrow 0} \frac{1}{\pi r^3} \cdot f(\xi, \eta, \zeta) \cdot \frac{4}{3}\pi r^3 = f(0,0,0) \cdot \frac{4}{3} = \frac{4}{3}.$$



投影法.

- 求由锥面 $z = \sqrt{x^2 + y^2}$, 平面 $y = x$, 柱面 $x^2 + y^2 = 2y$ 以及 xOy 面所围立体在第一卦限内的部分的体积.



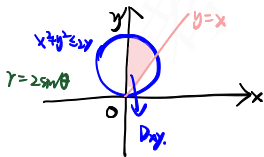
解: $\Omega = \{(x, y, z) \mid 0 \leq z \leq \sqrt{x^2 + y^2}, (x, y) \in D_{xy}\}$

$D_{xy} = \{(x, y) \mid 0 \leq x \leq y, x^2 + y^2 \leq 2y\} = \{(r, \theta) \mid 0 \leq r \leq 2\sin\theta, \frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}\}$

$$V = \iiint_{\Omega} 1 \, dv = \iint_{D_{xy}} dx \, dy \int_0^{\sqrt{x^2+y^2}} dz = \iint_{D_{xy}} \sqrt{x^2+y^2} \, dx \, dy$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^{2\sin\theta} r \cdot r \, dr = \frac{8}{3} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^3\theta \, d\theta$$

$$= -\frac{8}{3} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (1 - \cos^2\theta) d(\cos\theta) = \frac{10\sqrt{2}}{9}. \quad \text{〈计算留给大家〉}$$



字和分=字和分
截面积.

$$\iiint_{\Omega} f(x, y, z) dv = C$$

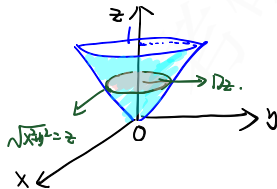
- 设函数 $f(x, y, z)$ 连续, 且

$$f(x, y, z) = \sqrt{x^2 + y^2} + z \iiint_{\Omega} f(x, y, z) dv, \text{ 其中区域}$$

$$\Omega = \{(x, y, z) | \sqrt{x^2 + y^2} \leq z \leq 1\}. \text{ 求 } f(x, y, z) \text{ 的表达式.}$$

解: 设 $\iiint_{\Omega} f(x, y, z) dv = C$, 则 $f(x, y, z) = \sqrt{x^2 + y^2} + C z = \sqrt{x^2 + y^2} + \frac{2\pi}{12-3\pi} z$.

在 Ω 上. 两边同时积分 $C = \iiint_{\Omega} \sqrt{x^2 + y^2} dv + C \cdot \iiint_{\Omega} z dv = \frac{\pi}{6} + \frac{\pi}{4} C \Rightarrow \frac{4-\pi}{4} C = \frac{\pi}{6} \Rightarrow C = \frac{2\pi}{12-3\pi}$.



$$D_z = \{(x, y) | x^2 + y^2 \leq z^2\}$$

$$\iiint_{\Omega} \sqrt{x^2 + y^2} dv = \int_0^1 dz \int_0^{2\pi} d\theta \int_0^z r \cdot r dr = 2\pi \int_0^1 \frac{z^3}{3} dz = 2\pi \times \frac{1}{12} = \frac{\pi}{6}.$$

$$\iiint_{\Omega} z dv = \int_0^1 z dz \int_{D_z} dx dy = \int_0^1 \pi z^3 dz = \frac{\pi}{4}.$$

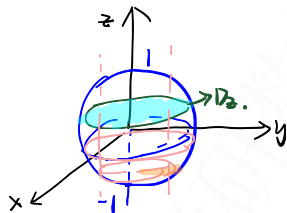
- 设 $f(z)$ 连续, $\Omega = \{(x, y, z) | x^2 + y^2 + z^2 \leq 1\}$. 证明

$$\iiint_{\Omega} f(z) dv = \pi \int_{-1}^1 f(z) (1 - z^2) dz.$$

常想型题: 球面坐标. 难.

$$x^2 + y^2 = 1 - z^2.$$

积的次序很关键.
最后才积 z .



直接法: 投影到 xy 面. 先积 xy 面, 再积 z .
球面坐标: 先积 r, θ , 再积 z .

解: $\Omega \rightarrow$ 平行于 xy 面截 Ω . 截面 $D_z = \{(x, y) | x^2 + y^2 \leq 1 - z^2\}$.

D_z 的面积 $\pi \cdot (1 - z^2)$

$$\iiint_{\Omega} f(z) dv = \int_{-1}^1 f(z) dz \int_{D_z} dx dy = \int_{-1}^1 \pi f(z) (1 - z^2) dz = \pi \int_{-1}^1 f(z) (1 - z^2) dz.$$

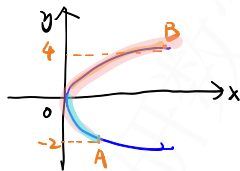
法二: $\Omega = \{(r, \theta, z) | 0 \leq \theta \leq 2\pi, 0 \leq r \leq \sqrt{1 - z^2}, -1 \leq z \leq 1\}$.

$$\iiint_{\Omega} f(z) dv = \int_{-1}^1 f(z) dz \int_0^{2\pi} d\theta \int_0^{\sqrt{1 - z^2}} r dr = 2\pi \int_{-1}^1 f(z) \cdot \frac{1 - z^2}{2} dz = \pi \int_{-1}^1 f(z) (1 - z^2) dz.$$

考点 II: 第一类曲线积分

参数化 化定积分

- 求 $\int_L y ds$, L 为抛物线 $y^2 = 2x$ 连接 $A(2, -2)$ 到 $B(8, 4)$ 的弧.



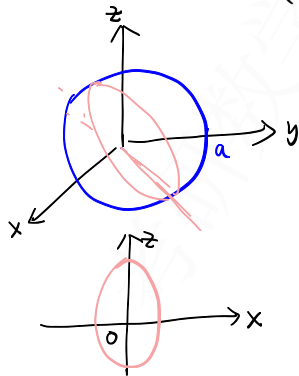
解: 选 y 作参数 $\begin{cases} x = \frac{y^2}{2} \\ y = y \end{cases} \quad -2 \leq y \leq 4. \quad ds = \sqrt{1+y^2} dy.$

$$\begin{aligned} \int_L y ds &= \int_{-2}^4 y \cdot \sqrt{1+y^2} dy = \frac{1}{2} \int_{-2}^4 \sqrt{1+y^2} d(1+y^2) \\ &= \frac{1}{2} \times \frac{2}{3} (1+y^2)^{\frac{3}{2}} \Big|_{-2}^4 \\ &= \frac{1}{3} (17\sqrt{17} - 5\sqrt{5}). \end{aligned}$$

找到 L 的在 zOx 面的投影曲线.

• 求 $\int_L x^2 ds$, $L: \begin{cases} x^2 + y^2 + z^2 = a^2, (a > 0). \\ x - y = 0. \end{cases}$

$$\begin{cases} x = \\ y = \\ z = \end{cases}$$

解: L 为一个大圆.需要写出 L 的参数方程.在 L 的方程中消去变量 y .

$$\text{得 } 2x^2 + z^2 = a^2$$

$$\frac{x^2}{\frac{a^2}{2}} + \frac{z^2}{a^2} = 1$$

投影曲面与 $\begin{cases} 2x^2 + z^2 = a^2 \\ y = 0. \end{cases}$ 即 L 在 zOx 面的投影曲线. L 的参数方程

$$\begin{cases} x = \frac{a}{\sqrt{2}} \cos t \\ y = \frac{a}{\sqrt{2}} \cos t \\ z = a \sin t \end{cases}$$

 $(x=y, x, z \text{ 由投影得来})$

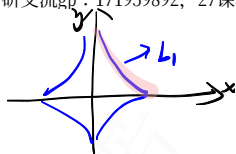
$$0 \leq t < 2\pi.$$

$$ds = \sqrt{\frac{a^2}{2} \sin^2 t + \frac{a^2}{2} \sin^2 t + a^2 \cos^2 t} dt = a dt.$$

$$\int_L x^2 ds = \int_0^{2\pi} \frac{a^2}{2} \cos^2 t \cdot a dt = \frac{a^3}{2} \int_0^{2\pi} \cos^2 t dt = \frac{a^3}{2} \int_0^{2\pi} \frac{1 + \cos 2t}{2} dt = \frac{a^3 \pi}{2}$$

星形线.

$$\cos^3 t + \sin^3 t = 1.$$



- 设曲线 L 为 $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1$, 求 L 的长度.

解: L 为星形线, 关于 x 轴, y 轴均对称. L 的长度为它在第一象限内的部分 L_1 的长度的4倍.

L_1 的参数方程:
$$\begin{cases} x = \cos^3 t \\ y = \sin^3 t \end{cases} \quad 0 \leq t \leq \frac{\pi}{2}$$

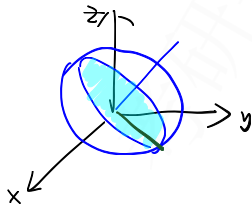
$$\begin{aligned} ds &= \sqrt{[3\cos^2 t \cdot (-\sin t)]^2 + (3\sin^2 t \cdot \cos t)^2} dt = \sqrt{9\cos^4 t \sin^2 t + 9\sin^4 t \cos^2 t} dt \\ &= 3\sqrt{\sin^2 t \cos^2 t (\cos^2 t + \sin^2 t)} dt = 3\sin t \cos t dt \end{aligned}$$

$$S = 4 \int_0^{\frac{\pi}{2}} 3 \sin t \cos t dt = 12 \int_0^{\frac{\pi}{2}} \sin t d(\cos t) = 12 \times \frac{1}{2} = 6$$

• 求 $\int_L (x + y^2) ds$, $L: \begin{cases} x^2 + y^2 + z^2 = a^2, \\ x + y + z = 0. \end{cases}$

轮换对称性 + 代曲线方程.

解: $\int_L (x + y^2) ds = \frac{1}{3} \int_L (x + y + z + x^2 + y^2 + z^2) ds \stackrel{\substack{x+y+z=0 \\ x^2+y^2+z^2=a^2}}{=} \frac{1}{3} a^2 \int_L ds = \frac{1}{3} a^2 \cdot 2\pi a = \frac{2\pi a^3}{3}$



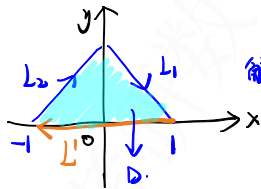
考点 III: 第二类曲线积分

积分曲线 L 关于 y 轴对称.

· 若 $f(x,y)$ 为关于 x 的奇函数, 则 $\int_L f(x,y) dx = 0$.

· 若 $f(x,y)$ 为关于 x 的偶函数, 则 $\int_L g(x,y) dy = 0$.

- 已知曲线 L 的方程为 $y = 1 - |x| (x \in [-1, 1])$, 起点是 $(-1, 0)$, 终点是 $(1, 0)$, 则曲线积分 $\int_L xy dx + x^2 dy = \underline{\hspace{2cm}}$.



L 关于 y 轴对称

利用参数方程硬算 留给大家

解: (法一) 对称性.

L 关于 y 轴对称. 考虑 x 的奇偶性.

xy 关于 x 奇 关于 x 积分. $\int_L xy dx = 0$

x^2 关于 x 偶 关于 y 积分. $\int_L x^2 dy = 0$

(法二). 格林公式.

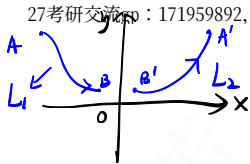
$L' + L$ 围成 D 的负向边界.

$L' = \begin{cases} x=x \\ y=0 \end{cases} \quad x \text{ 从 } 1 \text{ 到 } -1.$

$$\oint_{L'+L} xy dx + x^2 dy = - \iint_D (2x - x) dx dy = - \iint_D x dx dy = 0$$

$$\int_{L'} xy dx + x^2 dy = 0 \quad \text{原积分} = 0 - 0 = 0$$

1. 为 xoy 面上分段光滑曲线. 关于 y 轴对称.



• $f(x,y)$ 关于 x 奇, 则 $\int_L f(x,y) dx = 0$

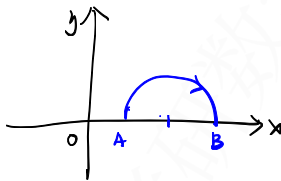
• $g(x,y)$ 关于 x 偶, 则 $\int_L f(x,y) dy = 0$.

$$L_1: \begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases} \quad t \text{ 从 } \alpha \text{ 到 } \beta \quad L_2: \begin{cases} x = -\varphi(t) \\ y = \psi(t) \end{cases} \quad t \text{ 从 } \beta \text{ 到 } \alpha$$

$$\begin{aligned} \int_L f(x,y) dx &= \int_{L_1} f(x,y) dx + \int_{L_2} f(x,y) dx = \int_{\alpha}^{\beta} f(\varphi(t), \psi(t)) \varphi'(t) dt + \int_{\beta}^{\alpha} \underbrace{f(-\varphi(t), \psi(t))}_{-f(\varphi(t), \psi(t))} (-\varphi'(t) dt) \\ &= \int_{\alpha}^{\beta} f(\varphi(t), \psi(t)) \varphi'(t) dt - \int_{\alpha}^{\beta} f(\varphi(t), \psi(t)) \varphi'(t) dt = 0 \end{aligned}$$

$$\begin{aligned} \int_L g(x,y) dy &= \int_{L_1} g(x,y) dy + \int_{L_2} g(x,y) dy = \int_{\alpha}^{\beta} \underbrace{g(\varphi(t), \psi(t))}_{g(\varphi(t), \psi(t))} \underbrace{\psi'(t) dt}_{\psi'(t) dt} + \int_{\beta}^{\alpha} \underbrace{g(-\varphi(t), \psi(t))}_{g(\varphi(t), \psi(t))} \underbrace{\psi'(t) dt}_{\psi'(t) dt} \\ &= \int_{\alpha}^{\beta} g(\varphi(t), \psi(t)) \psi'(t) dt - \int_{\alpha}^{\beta} g(\varphi(t), \psi(t)) \psi'(t) dt = 0. \end{aligned}$$

- 求 $\int_L (x+y)dx + (y-x)dy$, L 为圆周 $(x-2)^2 + y^2 = 1$ 的上半圆, 自点 $A(1,0)$ 到点 $B(3,0)$.



解: 初值 + 格林 留作练习

写出参数方程: $\begin{cases} x = 2 + \cos t \\ y = \sin t \end{cases}$

A 对应 π

B 对应 0.

t 从 π 到 0.

$$\int_L (x+y)dx + (y-x)dy$$

$$= \int_0^1 x dx$$

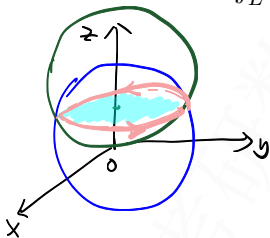
$$L: \begin{cases} x=x \\ y=0 \end{cases}$$

$$= \left. \frac{x^2}{2} \right|_0^1 = -4.$$

$$\oint_{L+L'} = \oint_L = \pi - (-4) = \pi + 4.$$

$$\begin{aligned} \int_L (x+y)dx + (y-x)dy &= \int_{\pi}^0 (2 + \cos t + \sin t)(-\sin t) + (\sin t - 2 - \cos t)(\cos t) dt \\ &= \int_{\pi}^0 (-2\sin t - 2\cos t - 1) dt = \int_0^{\pi} (2\sin t + 2\cos t + 1) dt \\ &= 4 + \pi. \end{aligned}$$

- 设闭曲线 $L: \begin{cases} x^2 + y^2 + z^2 = 1, \\ x^2 + y^2 + z^2 = 2z, \end{cases}$ 从 z 轴正向看去为逆时针方向, 求 $\oint_L (x+y)dx$.



解: $\begin{cases} x^2 + y^2 + z^2 = 1 \\ x^2 + y^2 + z^2 = 2z \end{cases} \Rightarrow z = \frac{1}{2}$ 代入 $x^2 + y^2 + z^2 = 1$ 得 $\begin{cases} x^2 + y^2 = \frac{3}{4} \\ z = \frac{1}{2} \end{cases}$

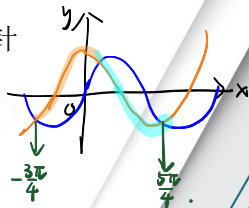
取参数方程 $\begin{cases} x = \frac{\sqrt{3}}{2} \cos t \\ y = \frac{\sqrt{3}}{2} \sin t \\ z = \frac{1}{2} \end{cases}$ 取 t 从 $-\pi$ 到 π .

$$\begin{aligned} \oint_L (x+y)dx &= \int_{-\pi}^{\pi} \frac{\sqrt{3}}{2} (\sin t + \cos t) \left(-\frac{\sqrt{3}}{2} \sin t\right) dt = -\frac{3}{4} \int_{-\pi}^{\pi} \sin^2 t dt = -\frac{3}{4} \int_{-\pi}^{\pi} \frac{1 - \cos 2t}{2} dt \\ &= -\frac{3}{4} \pi. \end{aligned}$$

$\cos t \cdot \sin t$ 关于 t 奇

- 设闭曲线 $L: \begin{cases} x^2 + y^2 + z^2 = 2, \\ x^2 + y^2 + z^2 = 2z, \end{cases}$
方向, 求 $\oint_L |y-x|dx + \underline{zdz}$.

从 z 轴正向看去为逆时针



解:

$$\begin{cases} x^2 + y^2 + z^2 = 2 \\ x^2 + y^2 + z^2 = 2z \end{cases} \Rightarrow z=1$$

代入 $x^2 + y^2 + z^2 = 2$
得 $x^2 + y^2 = 1$

$$L = \begin{cases} x = \cos t \\ y = \sin t \\ z = 1 \end{cases}$$

$$t \text{ 从 } -\frac{3\pi}{4} \text{ 到 } \frac{5\pi}{4}$$

$$\oint_L |y-x|dx + \underline{zdz} = \int_{-\frac{3\pi}{4}}^{\frac{5\pi}{4}} |\sin t - \cos t| (-\sin t) dt$$

$$= - \int_{-\frac{3\pi}{4}}^{\frac{5\pi}{4}} |\sqrt{2} \sin(t - \frac{\pi}{4})| \cdot \sin t \, dt \xrightarrow{u = t - \frac{\pi}{4}} - \int_{-\pi}^{\pi} |\sqrt{2} \sin u| \cdot \sin(u + \frac{\pi}{4}) \, du$$

$$= -\sqrt{2} \int_{-\pi}^{\pi} |\sin u| \sin(u + \frac{\pi}{4}) \, du = - \int_{-\pi}^{\pi} |\sin u| (\sin u + \cos u) \, du$$

$$\sin t - \cos t = \sqrt{2} \sin(t - \frac{\pi}{4})$$

$$\sin(u + \frac{\pi}{4}) = \frac{\sqrt{2}}{2} \sin u + \frac{\sqrt{2}}{2} \cos u$$

$$\stackrel{\text{对称性}}{=} -2 \int_0^{\pi} |\sin u| \cos u \, du = -2 \int_0^{\pi} \sin u \cos u \, du = -2 \int_0^{\pi} \sin u \, d(\sin u)$$

$$= 0.$$

考点 IV: 格林公式、积分与路径无关

- 设 $L_1: x^2 + y^2 = 1$, $L_2: x^2 + y^2 = 2$, $L_3: x^2 + 2y^2 = 2$,

$L_4: 2x^2 + y^2 = 2$ 为四条逆时针方向的平面曲线, 记

$$I_i = \oint_{L_i} \left(y + \frac{y^3}{6} \right) dx + \left(2x - \frac{x^3}{3} \right) dy \quad (i = 1, 2, 3, 4), \text{ 则}$$

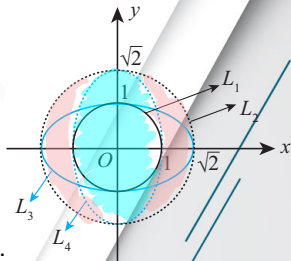
$$\max\{I_1, I_2, I_3, I_4\} = (\quad)$$

(A) I_1 .

(B) I_2 .

(C) I_3 .

(D) I_4 .



格林公式 = 重积分 < 被积函数相同, 积分区域不同 > L_i 围成 D_i .

解: $I_i = \iint_{D_i} \left(2 - x^2 - 1 - \frac{y^2}{2} \right) dx dy = \iint_{D_i} \left(1 - x^2 - \frac{y^2}{2} \right) dx dy$

分析被积函数的符号

$$1 - x^2 - \frac{y^2}{2} = \frac{1}{2}(2 - 2x^2 - y^2)$$

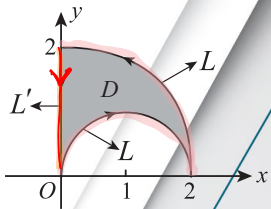
$2 - 2x^2 - y^2 = 0 \Rightarrow 2x^2 + y^2 = 2$. 当 (x, y) 在 D_4 内部时 $2x^2 + y^2 < 2$ 外 $2x^2 + y^2 > 2$.

$D_4 \subseteq D_2$. $I_2 - I_4 = \iint_{D_2 \setminus D_4} \left(1 - x^2 - \frac{y^2}{2} \right) dx dy < 0$

$D_1 \subseteq D_4$. $I_4 - I_1 = \iint_{D_4 \setminus D_1} \left(1 - x^2 - \frac{y^2}{2} \right) dx dy > 0$

$I_3 = \iint_{D_3 \cap D_4} + \iint_{D_3 \setminus D_4} < \iint_{D_4} = I_4$.

- 已知 L 是第一象限中从点 $(0,0)$ 沿圆周 $x^2 + y^2 = 2x$ 到点 $(2,0)$, 再沿圆周 $x^2 + y^2 = 4$ 到点 $(0,2)$ 的曲线段. 计算曲线积分 $I = \int_L 3x^2 y dx + (x^3 + x - 2y) dy$.



解 添加一条曲线 L' , 使之从点 $(0,2)$ 到点 $(0,0)$, 方向沿 y 轴负向. L 与 L' 围成一平面封闭区域 D , $L + L'$ 为区域 D 的正向边界.

$$\oint_{L+L'} 3x^2 y dx + (x^3 + x - 2y) dy = \iint_D (3x^2 + 1 - 3x^2) dx dy = \iint_D dx dy = \frac{1}{4} \cdot \pi \cdot 2^2 - \frac{1}{2} \cdot \pi \cdot 1^2 = \frac{\pi}{2}.$$

$$\int_{L'} 3x^2 y dx + (x^3 + x - 2y) dy = \int_2^0 (-2y) dy = \int_0^2 2y dy = 4.$$

$$L' = \begin{cases} x=0 \\ y=y \end{cases} \quad y \text{ 从 } 2 \text{ 到 } 0.$$

$$\text{原积分} = \frac{\pi}{2} - 4$$

补的曲线上的曲线积分一般都应该比较简单.

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0$$

- 设在上半平面 $D = \{(x, y) | y > 0\}$ 内, 函数 $f(x, y)$ 具有连续偏导数, 且对任意的 $t > 0$ 都有 $f(tx, ty) = t^{-2}f(x, y)$. 证明: 对 D 内的任意分段光滑的有向简单闭曲线 L , 都有

$$\oint_L yf(x, y)dx - xf(x, y)dy = 0.$$

证: 证明 $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0$.

证: 设 L 围成的区域为 G . 由格林公式

$$\begin{aligned} \oint_L yf(x, y)dx - xf(x, y)dy &= \iint_G [-f(x, y) - xf'_1(x, y) - f(x, y) - yf'_2(x, y)] dx dy \\ &= \iint_G [-2f(x, y) - xf'_1(x, y) - yf'_2(x, y)] dx dy. \end{aligned}$$

对 $f(tx, ty) = t^{-2} f(x, y)$ 关于 t 求导.

对任意 $t > 0$, 以及 $(x, y) \in D$, 均有.

$$x f'_1(tx, ty) + y f'_2(tx, ty) = -2t^{-3} f(x, y).$$

令 $t=1$ 则得 $x f'_1(x, y) + y f'_2(x, y) = -2f(x, y)$

↓

$$-2f(x, y) - x f'_1(x, y) - y f'_2(x, y) = 0.$$

则得: $\iint_G [-2f(x, y) - x f'_1(x, y) - y f'_2(x, y)] dx dy = 0.$

● 验证

$$\int_{(0,0)}^{(\frac{\pi}{2},1)} (2xy^3 - y^2 \cos x) dx + (1 - 2y \sin x + 3x^2 y^2) dy$$

与路径无关，并计算该积分。

验证积分与路径无关。

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0.$$

计算积分：

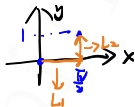
① 选择好算的路径 折线法。

② 找原函数。

解：

$$\begin{cases} \frac{\partial Q}{\partial x} = -2y \cos x + 6xy^2 \\ \frac{\partial P}{\partial y} = 6xy^2 - 2y \cos x \end{cases} \Rightarrow \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} \Rightarrow \text{积分与路径无关.}$$

验证 → 折线法：



$$L_1: \begin{cases} x = x \\ y = 0 \end{cases} \quad x \text{ 从 } 0 \text{ 到 } \frac{\pi}{2}$$

$$L_2: \begin{cases} x = \frac{\pi}{2} \\ y = y \end{cases} \quad y \text{ 从 } 0 \text{ 到 } 1.$$

$$\text{原积分} = \int_{L_1} + \int_{L_2}$$

$$= \int_0^{\frac{\pi}{2}} 0 \cdot dx + \int_0^1 (1 - 2y + 3 \cdot \frac{\pi^2}{4} \cdot y^2) dy = (y - y^2 + \frac{\pi^2}{4} y^3) \Big|_0^1 = \frac{\pi^2}{4}$$

<例2> 找原函数.

$$(2xy^3 - y^2 \cos x) dx + (1 - 2y \sin x + 3x^2 y^2) dy$$

$$= 2xy^3 dx + 3x^2 y^2 dy - (y^2 \cos x dx + 2y \sin x dy) + dy.$$

$$= d(x^2 y^3) - d(y^2 \sin x) + dy$$

$$= d(x^2 y^3 - y^2 \sin x + y).$$

$$\int_{(0,0)}^{(\frac{\pi}{2}, 1)} = (x^2 y^3 - y^2 \sin x + y) \Big|_{(0,0)}^{(\frac{\pi}{2}, 1)} = \frac{\pi^2}{4} - 0 = \frac{\pi^2}{4}.$$

- 设 $f(x)$ 有二阶连续导数, 且 $f(0) = 0$, $f'(0) = -1$. 已知曲线积分

$$\int_L [xe^{2x} - 6f(x)] \sin y dx - [5f(x) - f'(x)] \cos y dy$$

$$\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$$

与积分路径无关, 求 $f(x)$.

解 设 $P(x, y) = [xe^{2x} - 6f(x)] \sin y$,
 $Q(x, y) = -[5f(x) - f'(x)] \cos y$.

由积分与路径无关可得, $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, $\Rightarrow xe^{2x} - 6f(x) = f'(x) - 5f'(x) \Rightarrow f'(x) - 5f'(x) + 6f(x) = xe^{2x}$

$$\frac{\partial P}{\partial y} = [xe^{2x} - 6f(x)] \cos y \quad \frac{\partial Q}{\partial x} = -[5f'(x) - f'(x)] \cos y$$

$$f''(x) - 5f'(x) + 6f(x) = 0 \text{ 的特征方程为 } r^2 - 5r + 6 = 0 \Rightarrow \underline{r=2}, \underline{r=3}.$$

$$\text{通解: } y = \underline{C_1 e^{2x}} + \underline{C_2 e^{3x}}.$$

$$x e^{2x} \cdot \underline{e^{\lambda x}} \quad \lambda = 2 \quad \text{设 } y^* = (Ax + B) \cdot x \cdot e^{2x} = \underline{(Ax^2 + Bx) e^{2x}} \text{ 为原方程的一个特解.}$$

$$(y^*)' = (2Ax + B) e^{2x} + (2Ax^2 + 2Bx) e^{2x} = \underline{[2Ax^2 + (2A+2B)x + B] e^{2x}}.$$

$$\begin{aligned} (y^*)'' &= (4Ax + 2A + 2B) e^{2x} + [4Ax^2 + (4A + 4B)x + 2B] e^{2x} \\ &= [4Ax^2 + (8A + 4B)x + 2A + 4B] e^{2x}. \end{aligned}$$

$$\begin{aligned} A &= -\frac{1}{2} & B &= -1 \\ -x^2 - 3x - 1 \end{aligned}$$

$$\underline{\text{代入 } y'' - 5y' + 6y = x e^{2x}}.$$

$$[4Ax^2 + (8A + 4B)x + 2A + 4B - 10Ax^2 - (10A + 10B)x - 5B + 6Ax^2 + 6Bx] e^{2x} = x e^{2x}.$$

$$(-2Ax + 2A - B) e^{2x} = x e^{2x}.$$

$$A = -\frac{1}{2} \quad B = -1 \quad y^* = -\frac{x}{2}(x+2) e^{2x}$$

$$f(x) = C_1 e^{2x} + C_2 e^{3x} - \frac{x}{2}(x+2) e^{2x}$$

$$f(x) = 2C_1 e^{2x} + 3C_2 e^{3x} + (-x^2 - 3x - 1) e^{2x}$$

考点 V: 第一类曲面积分

$$f(0)=0 \Rightarrow c_1+c_2=0$$

$$f'(0)=-1 \Rightarrow 2c_1+3c_2-1=-1$$

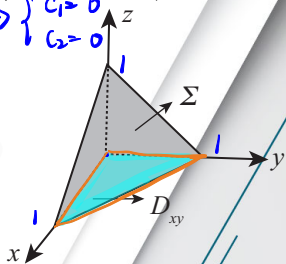
$$f(x) = -\frac{x}{2}(x+y)e^{2x}$$

$$\Rightarrow \begin{cases} c_1=0 \\ c_2=0 \end{cases}$$

转化为二重积分

- 设 $\Sigma = \{(x, y, z) | x + y + z = 1, x \geq 0, y \geq 0, z \geq 0\}$, 则

$$\iint_{\Sigma} y^2 dS = \underline{\hspace{2cm}}. \quad \text{全 } z=0 \quad x+y=1$$



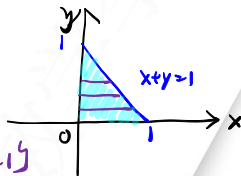
解 积分曲面满足方程 $z = 1 - x - y$. 计算 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$,

$$\frac{\partial z}{\partial x} = -1, \quad \frac{\partial z}{\partial y} = -1$$

$$dS = \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dx dy = \sqrt{3} dx dy$$

$$D_{xy} = \{(x, y) | x + y \leq 1, x \geq 0, y \geq 0\} \\ = \{(x, y) | 0 \leq x \leq 1-y, 0 \leq y \leq 1\}$$

$$\begin{aligned} \iint_{\Sigma} y^2 dS &= \iint_{D_{xy}} y^2 \cdot \sqrt{3} dx dy = \sqrt{3} \int_0^1 y^2 dy \int_0^{1-y} dx = \sqrt{3} \int_0^1 y^2 (1-y) dy \\ &= \sqrt{3} \int_0^1 (y^2 - y^3) dy = \sqrt{3} \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{\sqrt{3}}{12}. \end{aligned}$$



X型 先y后x.
Y型 先x后y

• 计算 $\iint_{\Sigma} x^2 dS$, 其中 Σ 为球心在原点的单位球面.

轮换对称性.

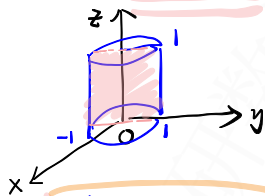
解: $\iint_{\Sigma} x^2 dS = \frac{1}{3} \iint_{\Sigma} (x^2 + y^2 + z^2) dS = \frac{1}{3} \iint_{\Sigma} dS = \frac{1}{3} \cdot 4\pi = \frac{4\pi}{3}.$

$x^2 + y^2 + z^2 = 1.$

- 求 $\iint_{\Sigma} \frac{ds}{x^2+y^2+z^2}$, 其中 Σ 为介于平面 $z=0$ 与 $z=1$ 之间的柱面 $x^2+y^2=1$.

zOx 面 or yOz 面.
 $\downarrow$
 $y = y(z, x)$ $x = x(y, z)$.

曲面分块.



解: 将 Σ 投影到 yOz 面. 将 Σ 分为 $\Sigma_{前}$, $\Sigma_{后}$. (在 yOz 面前/后).

$$\iint_{\Sigma} \frac{ds}{x^2+y^2+z^2} \stackrel{\text{代入}}{=} \iint_{\Sigma} \frac{ds}{1+z^2} = 2 \iint_{\Sigma_{前}} \frac{ds}{1+z^2}.$$

Σ 关于 yOz 面对称.

$\frac{1}{1+z^2}$ 为关于 x 的偶函数.

对称性.

$\Sigma_{前}$ 的参数 $x = \sqrt{1-y^2}$. $\frac{\partial x}{\partial y} = \frac{-y}{\sqrt{1-y^2}} = \frac{-y}{1-y^2}$. $\frac{\partial x}{\partial z} = 0$.

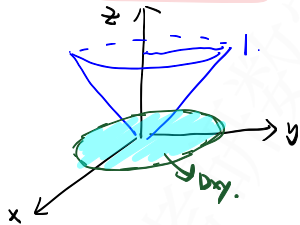
$D_{yz} = \{(y, z) | -1 \leq y \leq 1, 0 \leq z \leq 1\}$.

$ds = \sqrt{1 + \left(\frac{\partial x}{\partial y}\right)^2} dy dz = \frac{1}{\sqrt{1-y^2}} dy dz$.

$$\frac{y^2}{1-y^2} + 1 = \frac{1}{1-y^2}.$$

$$\begin{aligned}\sqrt{\text{面积}} &= 2 \iint_{\Sigma_{xy}} \frac{ds}{1+z^2} = 2 \int_0^1 \frac{1}{1+z^2} dz \int_{-1}^1 \frac{1}{\sqrt{1-y^2}} dy = 2 \cdot \arctan z \Big|_0^1 \cdot \arcsin y \Big|_{-1}^1 \\ &= 2 \times \frac{\pi}{4} \times \pi = \frac{\pi^2}{2}.\end{aligned}$$

- 计算 $I = \iint_{\Sigma} (x^2 + y^2 + z^2 + x + y + z) dS$, 其中 Σ 为 $x^2 + y^2 = z^2 (0 \leq z \leq 1)$ 的表面. 对称性



解: Σ 关于 yOz 和 xOz 面均对称.

x 为关于 x 的奇函数
 y 为关于 y 的奇函数

$$\iint_{\Sigma} x dS = 0,$$

$$\iint_{\Sigma} y dS = 0.$$

$$I = \iint_{\Sigma} (x^2 + y^2 + z^2 + z) dS.$$

$$= \iint_{D_{xy}} [z(x^2 + y^2) + \sqrt{x^2 + y^2}] \sqrt{2} dx dy$$

$$= \sqrt{2} \int_0^{2\pi} d\theta \int_0^1 (2r^2 + r) \cdot r dr$$

$$D_{xy} = \{(x, y) | x^2 + y^2 \leq 1\}$$

$$z = \sqrt{x^2 + y^2}.$$

$$\frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} \quad \frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$dS = \sqrt{1 + \frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2}} dx dy = \sqrt{2} dx dy.$$

$$= 2\sqrt{2}\pi \cdot \int_0^1 (2r^3 + r^2) dr$$

$$= 2\sqrt{2}\pi \left(\frac{r^4}{2} + \frac{r^3}{3} \right) \Big|_0^1 = \frac{5\sqrt{2}}{3}\pi.$$



- ↓ 实为梯度方向

$$\frac{\partial z}{\partial x} = 2ax \quad \frac{\partial z}{\partial y} = 2by \quad \text{梯度方向 } (2ax, 2by) \\ (6a, 6b)$$

$(6a, 8b)$ 与 $(-3, -4)$ 同向. $\frac{6a}{-3} = \frac{8b}{-4} > 0 \Rightarrow a=b < 0$

$$\sqrt{(6a)^2 + (8b)^2} = 10 \Rightarrow (6a)^2 + (8b)^2 = 100 \Rightarrow a^2 = 1 \Rightarrow a = -1 \quad b = -$$

(II) 由第 (I) 问可知曲面 Σ 为 $z = 2 - x^2 - y^2 (z \geq 0)$, 由第一类曲面积分的几何意义可知其面积等于 $\iint_{\Sigma} dS$.

$$\frac{\partial z}{\partial x} = -2x \quad \frac{\partial z}{\partial y} = -2y$$



令 $z = 0$, 可得 Σ 在 xOy 面的投影区域为

$$D_{xy} = \{(x, y) | x^2 + y^2 \leq 2\}.$$

$$\iint_{\Sigma} dS = \iint_{D_{xy}} \sqrt{1 + (z'_x)^2 + (z'_y)^2} dx dy = \iint_{D_{xy}} \sqrt{1 + 4x^2 + 4y^2} dx dy.$$

$$\begin{aligned} &\stackrel{\text{极坐标}}{=} \int_0^{2\pi} d\theta \int_0^{\sqrt{2}} \sqrt{1+4r^2} \cdot r dr = 2\pi \cdot \frac{1}{8} \int_0^{\sqrt{2}} \sqrt{1+4r^2} d(1+4r^2) \\ &= \frac{\pi}{4} \cdot \frac{2}{3} (1+4r^2)^{\frac{3}{2}} \Big|_0^{\sqrt{2}} = \frac{\pi}{6} \times (27 - 1) = \frac{13\pi}{3}. \end{aligned}$$

考点 VI: 第二类曲面积分

基础计算

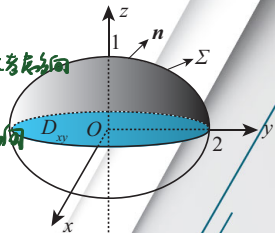
投影 投影面积.

定积分.

= 重积分 第二类 不考底面

ds.

第二类 考底面

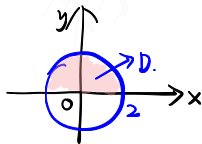


- 设 Σ 为曲面 $x^2 + y^2 + 4z^2 = 4 (z \geq 0)$ 的上侧, 则

$$\iint_{\Sigma} \sqrt{4 - x^2 - 4z^2} dx dy = \underline{\hspace{2cm}}.$$

解: $\iint_{\Sigma} \sqrt{4 - x^2 - 4z^2} dx dy = \iint_{\Sigma} \sqrt{y^2} dx dy = \iint_{\Sigma} |y| dx dy = \iint_{D_{xy}} |y| dx dy = \underline{\underline{2 \iint_{D_{xy}} y dx dy}}$

$$D_{xy} = \{(x, y) \mid x^2 + y^2 \leq 4\}$$



$$\begin{aligned} &= 2 \int_0^{\pi} d\theta \int_0^2 r \sin\theta \cdot r dr \\ &= 2 \int_0^{\pi} \sin\theta d\theta \int_0^2 r^2 dr \\ &= 2 \times 2 \times \frac{8}{3} = \frac{32}{3}. \end{aligned}$$

- 曲面 Σ 为锥面 $z^2 = x^2 + y^2 (0 \leq z \leq 1)$ 的下侧, 计算

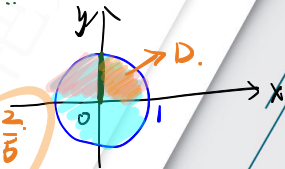
$$\iint_{\Sigma} |xyz| dx dy.$$



解 $\Sigma: z = \sqrt{x^2 + y^2} (0 \leq z \leq 1)$ 取下侧,

$$D_{xy} = \{(x, y) | x^2 + y^2 \leq 1\}.$$

$$\iint_{\Sigma} |xyz| dx dy = - \iint_{D_{xy}} |xy| \sqrt{x^2 + y^2} dx dy = - \iint_{D_{xy}} |xy| \sqrt{x^2 + y^2} dx dy = -\frac{2}{5}.$$



D_{xy} 关于 x 轴对称 也关于 y 轴对称. $|xy| \sqrt{x^2 + y^2}$ 既关于 y 偶, 又关于 x 偶

$$\begin{aligned} \iint_{D_{xy}} |xy| \sqrt{x^2 + y^2} dx dy &= 4 \iint_D |xy| \sqrt{x^2 + y^2} dx dy = 4 \iint_D xy \sqrt{x^2 + y^2} dx dy = 4 \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r^2 \sin\theta \cos\theta \cdot r \cdot r dr \\ &= 4 \int_0^{\frac{\pi}{2}} \sin\theta \cos\theta d\theta \int_0^1 r^4 dr = 4 \times \frac{1}{2} \times \frac{1}{5} = \frac{2}{5}. \end{aligned}$$

合一投影法. 西美曲面和它的投影.

$$\sin \theta d(\sin \theta)$$

- 计算 $\iint_{\Sigma} (x^2 + y^2) dz dx + z dx dy$, Σ 为锥面 $z = \sqrt{x^2 + y^2}$ 上满足 $x \geq 0, y \geq 0, z \leq 1$ 的那一部分的下侧.

解: 法一 合一投影法:

$$\langle F(x, y, z) = 0 \rangle$$

$$F(x, y, z) = \sqrt{x^2 + y^2} - z$$

$$dz dx = ds \cos \beta, \quad dx dy = ds \cdot \cos \gamma$$

曲面方程 $F(x, y, z) = 0$. 法向量 $\langle F'_x, F'_y, F'_z \rangle$.

$$dz dx = \frac{dx dy}{\cos \gamma} \cdot \cos \beta$$

$$\text{指向下方} \quad \left\langle \frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}}, -1 \right\rangle \Rightarrow \frac{\cos \beta}{\cos \gamma} = \frac{-y}{\sqrt{x^2 + y^2}}$$

$\langle \cos \alpha, \cos \beta, \cos \gamma \rangle$

$$z = \sqrt{x^2 + y^2}$$

$$D_{xy} = \{(x, y) \mid x^2 + y^2 \leq 1, x \geq 0, y \geq 0\}$$



$$\iint_{\Sigma} (x^2 + y^2) dz dx + z dx dy = \iint_{\Sigma} \left[(x^2 + y^2) \cdot \frac{-y}{\sqrt{x^2 + y^2}} + z \right] dx dy = - \iint_{D_{xy}} [\sqrt{x^2 + y^2} \cdot (-y) + \sqrt{x^2 + y^2}] dx dy$$

$$= - \iint_{D_{xy}} \sqrt{x^2 + y^2} (1 - y) dx dy = - \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r(1 - r \sin \theta) \cdot r dr$$

$$\begin{aligned}
 &= -\int_0^{\frac{\pi}{2}} d\theta \int_0^1 (r^2 - r^3 \sin\theta) dr = -\int_0^{\frac{\pi}{2}} d\theta \int_0^1 r^2 dr + \int_0^{\frac{\pi}{2}} \sin\theta d\theta \int_0^1 r^3 dr \\
 &= -\frac{\pi}{2} \times \frac{1}{3} + 1 \times \frac{1}{4} = \underline{\underline{\frac{1}{4} - \frac{\pi}{6}}}.
 \end{aligned}$$

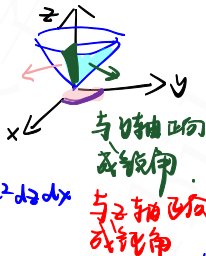
∠法 ⇒ 投影法.

$$\iint_{\Sigma} (x^2 + y^2) dz dx$$

代
 $x^2 + y^2 = z^2$

$$\iint_{\Sigma} z^2 dz dx = \int_{D_{zx}} z^2 dz dx$$

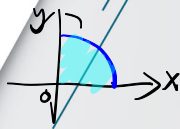
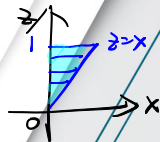
$$= \int_0^1 z^2 dz \int_0^z dx = \int_0^1 z^3 dz = \frac{1}{4}.$$



$$z = x^2 + y^2 = z^2 \Rightarrow \begin{cases} y=0 \\ x^2 = z^2 \end{cases} \Rightarrow z=x$$

$$D_{zx} = \{(x,z) \mid 0 \leq x \leq z, 0 \leq z \leq 1\}$$

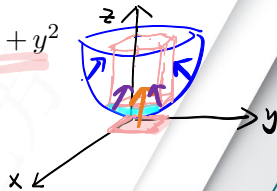
$$D_{xy} = \{(x,y) \mid x^2 + y^2 \leq 1, x \geq 0, y \geq 0\}$$



$$\iint_{\Sigma} z dx dy = -\iint_{D_{xy}} \sqrt{x^2 + y^2} dx dy = -\int_0^{\frac{\pi}{2}} d\theta \int_0^1 r \cdot r dr = -\frac{\pi}{2} \times \frac{1}{3} = -\frac{\pi}{6}.$$

$$\text{原积分} = \frac{1}{4} - \frac{\pi}{6}.$$

- 计算 $\iint_{\Sigma} e^y dydz + ye^x dzdx + x^2 y dx dy$, Σ 是抛物面 $z = x^2 + y^2$ 被平面 $x = 0, x = 1, y = 0, y = 1$ 所截部分的上侧.



解: 合-投影法.

$$F(x, y, z) = 0.$$

$$F(x, y, z) = z - x^2 - y^2$$

$$\begin{aligned} dydz &= ds \cos \alpha = dx dy \cdot \frac{\cos \alpha}{\cos \gamma} \\ dzdx &= ds \cos \beta = dx dy \cdot \frac{\cos \beta}{\cos \gamma} \\ dx dy &= ds \cos \gamma = dx dy \end{aligned}$$

$$D = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq 1\}$$

$$\langle F'_x, F'_y, F'_z \rangle = \langle -2x, -2y, 1 \rangle \Rightarrow \frac{\cos \alpha}{\cos \gamma} = -2x \quad \frac{\cos \beta}{\cos \gamma} = -2y.$$

$\downarrow$ $\downarrow$ $\downarrow$
 $\cos \alpha$ $\cos \beta$ $\cos \gamma$

$$\begin{aligned} \text{原式} &= \iint_D [e^y \cdot (-2x) + ye^x \cdot (-2y) + x^2 y] dx dy = \iint_D (-2xe^y - 2y^2 e^x + x^2 y) dx dy \\ &= \int_0^1 dx \int_0^1 (x^2 y - 2xe^y - 2e^x y^2) dy \end{aligned}$$

$$= \int_0^1 \left(\frac{x^2 y^2}{2} - \underline{2xy} - \frac{2}{3} y^3 e^x \right) \bigg|_{y=0}^{y=1} dx.$$

$$= \int_0^1 \left(\frac{x^2}{2} - 2ex - \frac{2}{3} e^x + 2x \right) dx$$

$$= \int_0^1 \left(\frac{x^2}{2} + 2(1-e)x - \frac{2}{3} e^x \right) dx$$

$$= \left(\frac{x^3}{6} + (1-e)x^2 - \frac{2}{3} e^x \right) \bigg|_0^1$$

$$= \frac{11-10e}{6}.$$

- 设有界区域 Ω 由平面 $2x + y + 2z = 2$ 与三个坐标平面围成, Σ 为 Ω 整个表面的外侧, 计算曲面积分

$$I = \iint_{\Sigma} (x^2 + 1) dydz - 2y dzdx + 3z dx dy.$$

$$z = \frac{2-x-y}{2}$$

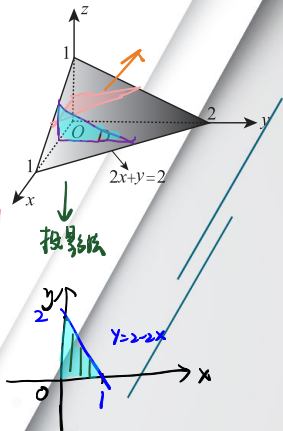
$$D_z = \{(x, y) | 2x + y \leq 2\}$$

解 由高斯公式可得,

$$I = \iiint_{\Omega} (2x - 2 + 3) dv = \iiint_{\Omega} 2x dv + \Omega \text{ 的体积 } V.$$

平面 $2x + y + 2z = 2$ 被三个坐标面所截得的有限部分在 xOy 面上的投影区域为 $D = \{(x, y) | 2x + y \leq 2\}$.

$$\Omega = \left\{ (x, y, z) | 0 \leq z \leq \frac{2 - 2x - y}{2}, (x, y) \in D \right\}.$$



采用“先单后重”的积分次序.

$$\begin{aligned}
 2 \iiint_{\Omega} x \, dv &= 2 \int_0^1 x \, dx \int_0^{2-2x} \int_0^{2-2x-y} dz = 2 \int_0^1 x \cdot \frac{2-2x-y}{2} \, dx \, dy = \int_0^1 x(2-2x-y) \, dx \, dy \\
 &= \int_0^1 x \, dx \int_0^{2-2x} (2-2x-y) \, dy = \int_0^1 x \left[(2-2x)y - \frac{y^2}{2} \right] \Big|_{y=0}^{y=2-2x} \, dx \\
 &= \int_0^1 x \left[(2-2x)^2 - \frac{(2-2x)^2}{2} \right] \, dx = 2 \int_0^1 x \cdot (1-x)^2 \, dx \\
 &= 2 \int_0^1 (x^3 - 2x^2 + x) \, dx = 2 \left(\frac{1}{4} - \frac{2}{3} + \frac{1}{2} \right) = \frac{1}{6}.
 \end{aligned}$$

$$\int_0^1 x(1-x)^2 \, dx \stackrel{u=1-x}{=} \int_1^0 (1-u)u^2 (-du) = \int_0^1 (u^2 - u^3) \, du = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

$$V = \frac{1}{3} \text{底面积} \times \text{高} = \frac{1}{3} \times \left(\frac{1}{2} \times 1 \times 1 \right) \times 1 = \frac{1}{6}.$$

$$I = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.$$

截面法 平行于 xoy 面截 Ω

$$D_z = \{(x, y) \mid 2x + y \leq 2 - 2z\}$$

$$\iiint_{\Omega} x \, dv = \int_0^1 dz \iint_{D_z} x \, dx \, dy$$

另外一种截面法: 平行于 yoz 面截 Ω . D_x

$$\iiint_{\Omega} x \, dv = \int_0^1 x \, dx \iint_{D_x} dy \, dz.$$

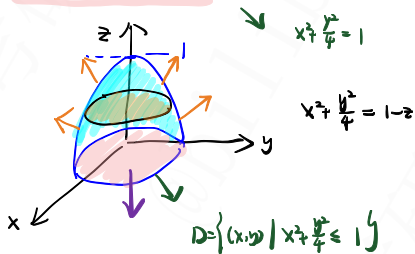
D_x 是半径为 $\sqrt{\frac{1-x^2}{3}}$ 的圆盘. 于是,

$$\Omega = \{(x, y, z) | (y, z) \in D_x, 0 \leq x \leq 1\}. \quad y^2 + z^2 = r^2.$$

$$\begin{aligned} \iiint_{\Omega} (1 + 3y^2 + 3z^2) dv &= \int_0^1 dx \underbrace{\iint_{D_x} (1 + 3y^2 + 3z^2) dy dz}_{\text{极坐标}} = \int_0^1 dx \int_0^{2\pi} d\theta \int_0^{\sqrt{\frac{1-x^2}{3}}} (1 + 3r^2) \cdot r dr \\ &= 2\pi \int_0^1 dx \int_0^{\sqrt{\frac{1-x^2}{3}}} (r + 3r^3) dr = 2\pi \int_0^1 \left(\frac{r^2}{2} + \frac{3r^4}{4} \right) \bigg|_0^{\sqrt{\frac{1-x^2}{3}}} dx \\ &= 2\pi \int_0^1 \left[\frac{1-x^2}{6} + \frac{1}{12} (1-x^2)^2 \right] dx = \frac{\pi}{6} \int_0^1 [2(1-x^2) + x^4 - 2x^2 + 1] dx \\ &= \frac{\pi}{6} \int_0^1 (x^4 - 4x^2 + 3) dx = \frac{\pi}{6} \left(\frac{1}{5} - \frac{4}{3} + 3 \right) = \frac{14\pi}{15}. \end{aligned}$$

$$I = \frac{14}{15}\pi - 0 = \frac{14}{15}\pi.$$

- 计算曲面积分 $I = \iint_{\Sigma} xzdydz + 2zydzdx + 3xydxdy$, 其中 Σ 为曲面 $z = 1 - x^2 - \frac{y^2}{4}$ ($0 \leq z \leq 1$) 的上侧.



解 曲面 Σ 为开口朝下的椭圆抛物面, 添加辅助平面

$\Sigma': z = 0(x^2 + \frac{y^2}{4} \leq 1)$, 并取其下侧, 则 Σ' 与 Σ 围成一封闭区域 Ω , 且法向量指向外侧.

对固定的 z , 平行于 xOy 面作 Ω 的横截面, 得 D_z . D_z 是一个椭圆域.

$$\Omega = \{(x, y, z) | (x, y) \in D_z, 0 \leq z \leq 1\}, \quad D_z = \left\{ (x, y) | x^2 + \frac{y^2}{4} \leq 1 - z \right\}.$$

利用高斯公式将曲面积分转化为三重积分, 并采用“先重后单”的积分次序.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{面积 } \pi ab$$

$$\frac{x^2}{1-z} + \frac{y^2}{4(1-z)} = 1$$

$$D_z \text{ 的面积 } \pi \cdot \sqrt{1-z} \cdot \sqrt{4(1-z)} = 2\pi(1-z).$$

$$\oiint_{\Sigma+\Sigma'} xzdydz + 2zydzdx + 3xydxdy = \iiint_{\Omega} (z+2z) dV = 3 \iiint_{\Omega} z dV.$$

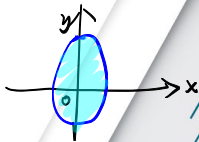
$$= 3 \int_0^1 z dz \oiint_{D_z} dx dy = 3 \int_0^1 z \cdot 2\pi(1-z) dz = 6\pi \int_0^1 z(1-z) dz$$

$$= 6\pi \cdot \left(\frac{z^2}{2} - \frac{z^3}{3}\right) \Big|_0^1 = \pi.$$

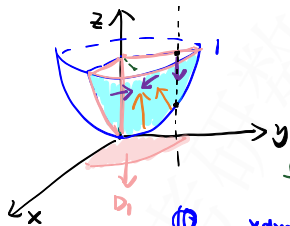
$$\oiint_{\Sigma'} \underline{xz} dy dz + 2zy \underline{dz} dx + 3xy \underline{dxdy} = \oiint_{\Sigma'} 3xy \underline{dxdy} = - \oiint_D 3xy \underline{dxdy} \stackrel{\text{对称性}}{=} 0.$$

Σ' 垂直于 Yoz 面, 与 zOx 面.

$$I = \pi - 0 = \pi.$$



- 计算 $I = \iint_{\Sigma} xdydz + ydzdx + zdxdy$, 其中 Σ 为 $z = x^2 + y^2$ 在第一卦限中 $0 \leq z \leq 1$ 的部分的上侧.



解: 添加辅助平面, $\Sigma_1: z=1$ ($x^2+y^2 \leq 1, x \geq 0, y \geq 0$)

$\Sigma_2: x=0$ (由 $z=y^2$, z 轴, $z=1$ 围成区域) (yoz面)

$\Sigma_3: y=0$ (由 $z=x^2$, z 轴, $z=1$ 围成区域) (xoz面)

$$\Omega = \{(r, \theta, z) | r^2 \leq z \leq 1, 0 \leq r \leq 1, 0 \leq \theta \leq \frac{\pi}{2}\}$$

$$\Sigma = \Sigma_1 + \Sigma_2 + \Sigma_3$$

$$x dy dz + y dz dx + z dx dy = -z \iiint_{\Omega} dr \quad \text{柱面坐标} \quad = -3 \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r dr \int_{r^2}^1 dz.$$

$$= -\frac{3\pi}{2} \int_0^1 r(1-r^2) dr = -\frac{3\pi}{2} \cdot \left(\frac{r^2}{2} - \frac{r^4}{4} \right) \Big|_0^1 = -\frac{3\pi}{2} \times \left(\frac{1}{2} - \frac{1}{4} \right) = -\frac{3\pi}{8}.$$

$$\iint_{\Sigma_1} \underline{x dy dz} + \underline{y dz dx} + \underline{z dx dy} = \iint_{\Sigma_1} dx dy = - \iint_{D_1} dx dy = -\frac{\pi}{4}.$$

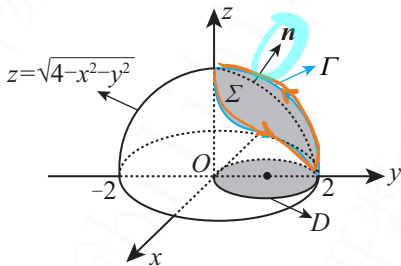
$$\iint_{\Sigma_2} \underline{x dy dz} + \underline{y dz dx} + \underline{z dx dy} = \iint_{\Sigma_2} 0 dy dz = 0.$$

$$\iint_{\Sigma_3} \underline{x dy dz} + \underline{y dz dx} + \underline{z dx dy} = \iint_{\Sigma_3} 0 dz dx = 0.$$

$$I = -\frac{3\pi}{8} - (-\frac{\pi}{4}) - 0 - 0 = -\frac{\pi}{8}.$$

斯托克斯公式.

- 计算曲线积分 $I = \oint_{\Gamma} z^2 dx + 2xy dy + 2yz dz$, 其中 Γ 是上半球面 $z = \sqrt{4 - x^2 - y^2}$ 与柱面 $x^2 + y^2 = 2y$ 的交线. 从 z 轴正向看去, Γ 沿逆时针方向.



斯托克斯公式

↓
曲面法向

↓
含参数

xy平面上的二重积分

解: 记 Σ 为上半球面上被 Γ 截下的部分. Σ 满足 $z = \sqrt{4-x^2-y^2}$.

Σ 的法向量 n 指向上侧. 该方向与 Γ 的定向符合右规则.

$$n = \langle -z'_x, -z'_y, 1 \rangle = \left\langle \frac{x}{\sqrt{4-x^2-y^2}}, \frac{y}{\sqrt{4-x^2-y^2}}, 1 \right\rangle.$$

$$\frac{\cos \alpha}{\cos \gamma} = \frac{x}{\sqrt{4-x^2-y^2}} \quad \frac{\cos \beta}{\cos \gamma} = \frac{y}{\sqrt{4-x^2-y^2}}$$

$$\langle \cos \alpha, \cos \beta, \cos \gamma \rangle$$

$$dydz = ds \cos \alpha = \frac{dx dy}{\cos \gamma} \cdot \cos \alpha$$

$$dzdx = ds \cos \beta = \frac{dx dy}{\cos \gamma} \cdot \cos \beta$$

$$dxdy = ds \cos \gamma.$$

$$I = \iint_{\Sigma} \begin{vmatrix} dydz & dzdx & dxdy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z^2 & 2xy & 2yz \end{vmatrix}$$

$$= \iint_{\Sigma} \underline{2z} dydz + \underline{2z} dzdx + \underline{2y} dxdy.$$

$$= \iint_{\Sigma} \left[2z \cdot \frac{\cos \alpha}{\cos \gamma} + 2z \cdot \frac{\cos \beta}{\cos \gamma} + 2y \right] dxdy$$

$$= \iint_{\Sigma} \left[2z \cdot \frac{x}{\sqrt{4-x^2-y^2}} + 2z \cdot \frac{y}{\sqrt{4-x^2-y^2}} + 2y \right] dxdy$$

$$= \iint_{\Sigma} (2x + 4y) dxdy = \iint_D (2x + 4y) dxdy$$

$$D = \{(x, y) \mid x^2 + y^2 \leq 2y\}$$

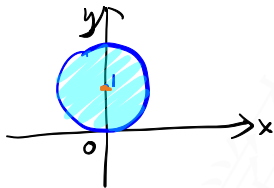
D 关于 y 轴对称.

$$= 4 \iint_D y dxdy$$

于是

$$\bar{y} = \frac{\iint_D y dxdy}{\iint_D dxdy}$$

$$\Rightarrow \iint_D y dxdy = \iint_D dxdy = \pi$$

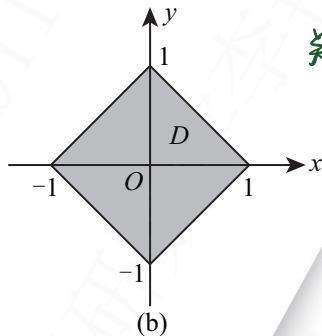
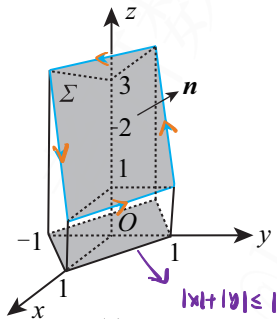
$C(x, y) = \langle 0, 1 \rangle$ 

$$= 4\pi.$$

$$\begin{aligned} \iint_D y \, dx \, dy &= \int_0^\pi d\theta \int_0^{2\sin\theta} r \sin\theta \cdot r \, dr \\ &= \int_0^\pi \sin\theta \, d\theta \int_0^{2\sin\theta} r^2 \, dr \\ &= \frac{8}{3} \int_0^\pi \sin\theta \cdot \sin^3\theta \, d\theta \\ &= \frac{8}{3} \int_0^\pi \sin^4\theta \, d\theta \\ &= \frac{8}{3} \times 2 \times \int_0^{\frac{\pi}{2}} \sin^4\theta \, d\theta \\ &= \frac{16}{3} \times \frac{\pi}{2} \times \frac{3}{4} \times \frac{1}{2} = \pi. \end{aligned}$$

$$I = \iint_D \begin{vmatrix} \cos\alpha & \cos\beta & \cos\gamma \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & z & y \end{vmatrix} dS$$

- 计算 $I = \oint_L (y^2 - z^2)dx + (2z^2 - x^2)dy + (3x^2 - y^2)dz$, 其中 L 是平面 $x + y + z = 2$ 与柱面 $|x| + |y| = 1$ 的交线, 从 z 轴正向看去, L 为逆时针方向.



关于 x 轴对称,
 y 轴

$$z = 2 - x - y$$

解 记平面 $x + y + z = 2$ 上由 L 所截得部分为 Σ .

根据右手规则, Σ 的法向量与 z 轴正向成锐角.

$$\vec{n} = (1, 1, 1) \quad (\cos \alpha, \cos \beta, \cos \gamma) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right).$$

$$ds = \sqrt{1 + (z'_x)^2 + (z'_y)^2} dx dy = \sqrt{3} dx dy.$$

$$D = \{(x, y) \mid |x| + |y| \leq 1\}. \quad D \text{ 是边长 } \sqrt{2} \text{ 的正方形, 面积为 } 2.$$

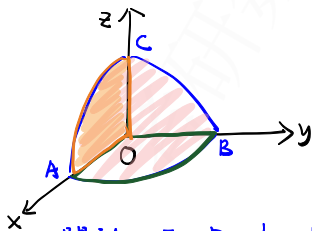
$$I = \iint_{\Sigma} \begin{vmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix} dS = \frac{1}{\sqrt{3}} \iint_{\Sigma} (-2y - 4z - \langle 6x + 2z \rangle + (-2x - 2y)) dS$$

$$= -\frac{2}{\sqrt{3}} \iint_{\Sigma} (4x + 2y + 3z) dS = -\frac{2}{\sqrt{3}} \iint_{\Sigma} (4x + 2y + 3(2 - x - y)) dS$$

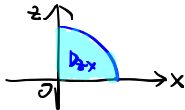
$$= -\frac{2}{\sqrt{3}} \iint_{\Sigma} (6 + x - y) dS = -2 \iint_D (6 + x - y) dx dy = -12 \iint_D dx dy = -24.$$

考点 VIII: 多元积分的应用

- 在球面 $x^2 + y^2 + z^2 = 1$ 上取以 $A(1, 0, 0)$, $B(0, 1, 0)$, $C(0, 0, 1)$ 为顶点的球面三角形 Σ , 如果该球面三角形的面密度为 $\rho = x^2 + z^2$, 则此球面三角形的质量 $m = \underline{\hspace{2cm}}$.



投影到 zox 面. $D_{zx} = \{(x, z) | x^2 + z^2 \leq 1, x \geq 0, z \geq 0\}$



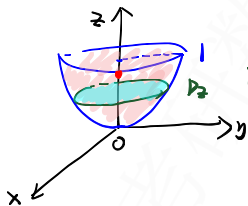
$$\begin{aligned} \text{解: } m &= \iint_{\Sigma} (x^2 + z^2) dS = \iint_{D_{zx}} (x^2 + z^2) \cdot \frac{1}{\sqrt{1-x^2-z^2}} dxdz = \int_0^{\frac{\pi}{2}} d\theta \int_0^1 \frac{r^2}{\sqrt{1-r^2}} \cdot r dr \\ r = \sin t &= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \frac{\sin^4 t}{\cos t} \cdot \cos t dt = \frac{\pi}{2} \cdot \int_0^{\frac{\pi}{2}} \sin^4 t dt = \frac{\pi}{2} \times \frac{2}{3} = \frac{\pi}{3}. \end{aligned}$$

$$\begin{aligned} y &= \sqrt{1-x^2-z^2} & dS &= \sqrt{1+(y'_x)^2+(y'_z)^2} dxdz \\ & & &= \sqrt{\frac{1}{1-x^2-z^2}} dxdz \\ y'_x &= \frac{-x}{\sqrt{1-x^2-z^2}} & y'_z &= \frac{-z}{\sqrt{1-x^2-z^2}} \end{aligned}$$

$$\frac{x^2}{1-x^2-z^2} + \frac{z^2}{1-x^2-z^2} + \frac{1-x^2-z^2}{1-x^2-z^2}$$

- 设 $\Omega = \{(x, y, z) | x^2 + y^2 \leq z \leq 1\}$, 则 Ω 的形心的竖坐标 $\bar{z} =$ _____.

解 由形心坐标公式知 $\bar{z} = \frac{\iiint_{\Omega} z dv}{\iiint_{\Omega} dv}$.



$$\Omega = \{(x, y) | x^2 + y^2 \leq z\}$$

Ω 的面积是 πz

$$\iint_{\Omega} dx dy = \pi z$$

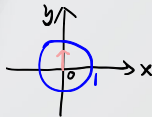
$$\bar{z} = \frac{1/10 \pi / 1}{1/10 \pi / 1} = \frac{2}{3}$$

$$\Omega = \{(x, y, z) | x^2 + y^2 \leq z \leq 1\}$$

$$\begin{aligned} \iint_{\Omega} z dv &= \int_0^1 z dz \iint_{\Omega} dx dy = \int_0^1 z \cdot \pi z dz \\ &= \frac{1}{3} \pi. \end{aligned}$$

$$\begin{aligned} \iint_{\Omega} dv &= \int_0^1 dz \iint_{\Omega} dx dy = \int_0^1 \pi z dz \\ &= \frac{1}{2} \pi. \end{aligned}$$

$$f(x) = f(x)$$



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$$(x)^2 + (y-z)^2 = (1-z)^2$$

||
x²

看底: $z=0: x^2+y^2=1$

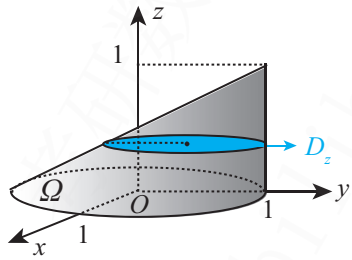
看截面: $z=z$ 与 $x^2+(y-z)^2=(1-z)^2$
 $\frac{1}{2}, \frac{1}{3}$ $x^2+(y-\frac{1}{2})^2=(1-\frac{1}{2})^2$

- 设 Ω 是由锥面 $x^2 + (y-z)^2 = (1-z)^2 (0 \leq z \leq 1)$ 与平面 $z=0$ 围成的锥体, 求 Ω 的形心坐标. $\langle \bar{x}, \bar{y}, \bar{z} \rangle$.

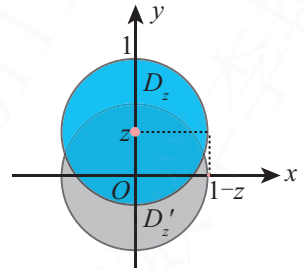
$$(0, \frac{1}{2}) \quad \frac{1}{2}$$

$$x^2+(y-\frac{1}{2})^2=(1-\frac{1}{2})^2$$

$$(0, \frac{1}{3}) \quad \frac{2}{3}$$



(a)



(b)

$$\iiint_{\Omega} y \, dv$$

$$\int_0^1 z \, dz$$

$$V = \pi \times 1 \times \frac{1}{3} = \frac{\pi}{3}$$

解 由于 Ω 关于 yOz 面对称, 故由对称性可知, $\bar{x} = 0$.

令 $z = 0$, 可得锥面与 xOy 面的交线为 $x^2 + y^2 = 1$,

沿平行于 xOy 面的方向作 Ω 的截面, 即以平面 $Z = z$ 截 Ω , 所得截面记为 D_z .

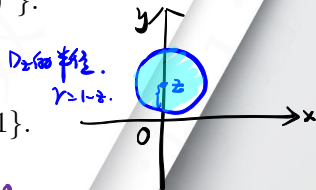
$$D_z = \{(x, y) | x^2 + (y - z)^2 \leq (1 - z)^2\}.$$

于是,

$$\Omega = \{(x, y, z) | (x, y) \in D_z, 0 \leq z \leq 1\}.$$

$$\begin{cases} x = r \cos \theta \\ y = z + r \sin \theta \end{cases}$$

$$dx dy = r dr d\theta.$$



利用“先重后单”的次序计算积分.

$$\begin{aligned}
 \iiint_{\Omega} y \, dz &= \int_0^1 dz \iint_{D_z} y \, dx \, dy = \int_0^1 dz \int_0^{2\pi} d\theta \int_0^{1-z} (\underline{z} + r \sin \theta) \cdot r \, dr \\
 &= \int_0^1 dz \int_0^{2\pi} \left[\left(\frac{z}{2} \cdot r^2 + \sin \theta \cdot \frac{r^3}{3} \right) \right]_{r=0}^{r=1-z} d\theta = \int_0^1 dz \int_0^{2\pi} \left[\frac{z}{2} \cdot (1-z)^2 + \sin \theta \cdot \frac{(1-z)^3}{3} \right] d\theta \\
 &= \int_0^1 2\pi \cdot \frac{z}{2} (1-z)^2 \, dz = \pi \int_0^1 z (1-z)^2 \, dz = \frac{\pi}{12}.
 \end{aligned}$$

$$\iiint_{\Omega} z \, dz = \int_0^1 z \, dz \iint_{D_z} dx \, dy = \int_0^1 z \cdot \pi (1-z)^2 \, dz = \frac{\pi}{12}.$$

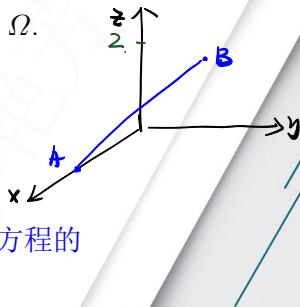
$$\bar{y} = \frac{\frac{\pi}{12}}{\frac{\pi}{4}} = \frac{1}{4}, \quad \bar{z} = \frac{\frac{\pi}{12}}{\frac{\pi}{4}} = \frac{1}{4}.$$

重心坐标 $(0, \frac{1}{4}, \frac{1}{4})$.

- 设直线 L 过 $A(1,0,0)$, $B(0,1,1)$ 两点, 将 L 绕 z 轴旋转一周得到曲面 Σ , Σ 与平面 $z=0$, $z=2$ 所围成的立体为 Ω .

(I) 求曲面 Σ 的方程;

(II) 求 Ω 的形心坐标.



空间曲线 $\Gamma: \begin{cases} F(x, y, z) = 0, \\ G(x, y, z) = 0 \end{cases}$ 绕 z 轴旋转所得旋转曲面方程的

求法为: 先将曲线 Γ 写成 $\begin{cases} x = x(z), \\ y = y(z) \end{cases}$ 的形式, 从而所求旋转曲面的方程为: $x^2 + y^2 = x^2(z) + y^2(z)$.

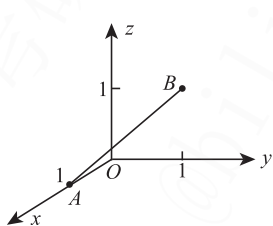
用坐标法求曲线方程

解 (I) 由于直线 L 过点 $A(1,0,0)$, $B(0,1,1)$, 故直线 L 的方向向量可取为 $\langle 1, -1, -1 \rangle$.

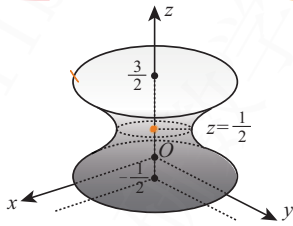
L 的方程为 $\frac{x-1}{1} = \frac{y}{-1} = \frac{z}{-1}$ $1-x=z \Rightarrow \begin{cases} x=1-z \\ y=z \end{cases}$

旋转曲面 Ω 的方程为 $x^2 + y^2 = (1-z)^2 + z^2$.

$\downarrow$
 $x^2 + y^2 - 2z^2 + 2z - 1 = 0 \Rightarrow x^2 + y^2 - 2(z - \frac{1}{2})^2 = \frac{1}{2}$ 单叶双曲面



(a)



(b)

Ω 关于 YOZ 面, ZOX 面都对称

$\bar{x}, \bar{y} = 0$

$D_z = \{(x,y) \mid x^2 + y^2 \leq 2z^2 - 2z + 1\}$

(II) 曲面方程为 $x^2 + y^2 - 2(z - \frac{1}{2})^2 = \frac{1}{2}$. 这是以点 $(0, 0, \frac{1}{2})$ 为中心, z 轴为对称轴的旋转单叶双曲面.

由于 Ω 关于 YOZ 面, 关于 ZOX 面均对称, 故 $\bar{x} = \bar{y} = 0$.

$$\begin{aligned} \iint_{\Omega} z \, dv &= \int_0^2 z \, dz \iint_{D_z} dx \, dy = \int_0^2 z \cdot \pi(2z^2 - 2z + 1) \, dz = \pi \int_0^2 (2z^3 - 2z^2 + z) \, dz \\ &= \pi \left(\frac{2z^4}{4} - \frac{2z^3}{3} + \frac{z^2}{2} \right) \Big|_0^2 = \pi \left(8 - \frac{16}{3} + 2 \right) = \frac{14\pi}{3}. \end{aligned}$$

$$\begin{aligned} \iint_{\Omega} 1 \, dv &= \int_0^2 dz \iint_{D_z} dx \, dy = \int_0^2 \pi(2z^2 - 2z + 1) \, dz = \pi \left(\frac{2z^3}{3} - z^2 + z \right) \Big|_0^2 \\ &= \pi \left(\frac{16}{3} - 4 + 2 \right) = \frac{10\pi}{3}. \end{aligned}$$

$$\bar{z} = \frac{14\pi/3}{10\pi/3} = \frac{7}{5}, \quad (\bar{x}, \bar{y}, \bar{z}) = (0, 0, \frac{7}{5}).$$

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 $\overrightarrow{OM}$

$$d\vec{r} = (dx, dy, dz)$$

- 在变力 $\mathbf{F} = yz\mathbf{i} + zx\mathbf{j} + xy\mathbf{k}$ 的作用下, 质点由原点沿直线运动到椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 上第一卦限的点 $M(\xi, \eta, \zeta)$, 问 ξ, η, ζ 取何值时, 力 $\mathbf{F}$ 所作的功 W 最大? 并求出 W 的最大值.

$$\overrightarrow{OM}: \begin{cases} x = \xi t \\ y = \eta t \\ z = \zeta t \end{cases} \quad t \in [0, 1]$$

$$\begin{aligned} \text{解: } W &= \int_{\overrightarrow{OM}} \vec{F} \cdot d\vec{r} = \int_{\overrightarrow{OM}} yz dx + zx dy + xy dz \\ &= 3 \int_0^1 \xi \eta \zeta t^2 dt = \xi \eta \zeta. \end{aligned}$$



用拉格朗日乘数法.

$$L(\xi, \eta, \zeta, \lambda) = \xi \eta \zeta + \lambda \left[1 - \frac{\xi^2}{a^2} - \frac{\eta^2}{b^2} - \frac{\zeta^2}{c^2} \right]$$

$$\begin{cases} L'_\xi = \eta \zeta - \frac{2\lambda \xi}{a^2} = 0 \\ L'_\eta = \xi \zeta - \frac{2\lambda \eta}{b^2} = 0 \end{cases}$$

$$\Rightarrow \xi \eta \zeta - 2\lambda \frac{\xi^2}{a^2} = 0$$

$$\Rightarrow \xi \eta \zeta - 2\lambda \frac{\eta^2}{b^2} = 0$$

$$\left\{ \begin{array}{l} L'_\xi = \xi\eta - \frac{2\lambda}{c^2} = 0 \\ L'_\lambda = 1 - \frac{\xi^2}{a^2} - \frac{\eta^2}{b^2} - \frac{\zeta^2}{c^2} = 0 \end{array} \right. \Rightarrow \xi\eta\zeta - 2\lambda \frac{\zeta^2}{c^2} = 0 \Rightarrow \xi\eta\zeta = 2\lambda \frac{\xi^2}{a^2} = 2\lambda \frac{\eta^2}{b^2} = 2\lambda \frac{\zeta^2}{c^2}$$

(1) 若 $\lambda \neq 0$, 则 $\frac{\xi^2}{a^2} = \frac{\eta^2}{b^2} = \frac{\zeta^2}{c^2}$ 再由 $\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} + \frac{\zeta^2}{c^2} = 1$ 得 $\frac{\xi^2}{a^2} = \frac{\eta^2}{b^2} = \frac{\zeta^2}{c^2} = \frac{1}{3}$.

$\xi = \frac{a}{\sqrt{3}}, \eta = \frac{b}{\sqrt{3}}, \zeta = \frac{c}{\sqrt{3}}$. 此时, $\xi\eta\zeta = \frac{abc}{3\sqrt{3}} = \frac{\sqrt{3}abc}{9}$.

(2) $\lambda = 0$ $\left\{ \begin{array}{l} \eta\zeta = 0, \\ \xi\zeta = 0 \\ \xi\eta = 0 \end{array} \right. \Rightarrow \xi, \eta, \zeta \text{ 至少有一个为零} \quad \xi\eta\zeta = \underline{\underline{0}}$

$W_{\max} = \frac{\sqrt{3}}{9} abc.$